

BILLAL HOSSAIN SHISHIR

ICT | Data Science and Analytics | Artificial Intelligence
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PROFESSIONAL SUMMARY

Final-year Information and Communication Technology (ICT) student majoring in Artificial Intelligence, with hands-on experience in Python, SQL, data engineering, machine learning, IoT monitoring, backend development and testing. Built a [portfolio](#) of five interactive project demos, including systems that process sensor data, detect anomalies and present operational insights. Undertook the role of Project Manager for a five-student capstone project team, coordinating requirements analysis, software development and testing, documentation development, and client consultation. Customer service experience in busy retail and hospitality environments.

EDUCATION

Bachelor of Information and Communication Technology (Artificial Intelligence)

University of Tasmania | Hobart, TAS | Expected completion November 2026

International First Year Diploma Science

University of Tasmania | Hobart, TAS | 2024

CORE TECHNICAL SKILLS

Programming and Data: Python, SQL, JavaScript, C, pandas, NumPy, data cleaning, transformation, analytics

Machine Learning and Monitoring: scikit-learn, classification, TF-IDF, anomaly detection, Isolation Forest, rolling statistics, operational metrics

APIs, Databases and IoT: FastAPI, REST APIs, SQLAlchemy, Pydantic, PostgreSQL, SQLite, MQTT, Chart.js

Cloud, DevOps and Testing: Docker, Docker Compose, GitHub Actions, pytest, Git/GitHub, Nectar Cloud, AWS fundamentals, Linux, Windows

Delivery and Collaboration: requirements analysis, functional testing, documentation, stakeholder communication, project coordination, presentations

PROFESSIONAL EXPERIENCE

Customer Service Representative | Bennett's Petroleum

Hobart, TAS | October 2024 - Present

- Provide customer service, handle transactions and resolve enquiries accurately in a fast-paced environment.
- Complete daily operational tasks independently, identify recurring issues and contribute practical solutions.

Crew Member | McDonald's Rosny Park

Hobart, TAS | June 2024 - September 2024

- Processed orders and transactions accurately and coordinated with colleagues during busy service periods.

Customer Service Assistant | Gadget and Gear

Dhaka, Bangladesh | April 2023 - December 2023

- Supported technology product enquiries, recommendations, payments, returns, complaints and stock tasks.

INDUSTRY ENGAGEMENT

Capstone Development Project - RAMIFY OS - Role Project Manager

University Industry Project | Hobart, TAS | July 2026 - Present

An AI-assisted product verification system designed to help consumers assess product authenticity before purchase by checking third-party laboratory reports and other supporting evidence.

- Lead a team of five students, coordinating requirements, development, testing, documentation and delivery.
- Work directly with the industry client to turn feedback into testable changes, priorities and team actions.
- Coordinate architecture material, user guidance, demonstrations and test evidence across multiple scenarios.

Open Innovation Hackathon | TasNetworks x University of Tasmania x AWS

Hobart, TAS | May 2026

- Collaborated on NetCab++, a prototype combining network records, GIS/design-sheet evidence and structured comparison for a TasNetworks asset-data challenge to support infrastructure planning decisions.

DataEngBytes Melbourne 2026 | Industry Conference

Melbourne, VIC | 23-24 July 2026

- Attended sessions on data and AI engineering, Snowflake, synthetic data, observability, data architecture and agent workflows.

SELECTED DATA ICT AND AI PROJECTS

AI Helpdesk Copilot

AI / Full-Stack / IT Support | [Live demo](#) | [GitHub](#)

- Built a FastAPI and SQLAlchemy helpdesk with PostgreSQL support, a TF-IDF and Logistic Regression classifier covering 7 support categories, transparent priority rules and retrieval over 70 troubleshooting articles.
- Implemented ticket creation, technician notes, similar-incident search, resolution analytics and 25 automated tests. Published an interactive GitHub Pages demo with the full implementation in the repository.

Real-Time Data Engineering Pipeline

Data Engineering / IoT | [Live demo](#) | [GitHub](#)

- Built a pipeline for 10 simulated IoT devices, ingesting temperature, humidity and energy events through MQTT, validating and enriching records, storing results in PostgreSQL and handling failed records.
- Implemented rule-based, rolling 3-sigma and Isolation Forest anomaly detection, operational metrics, latency and throughput monitoring, automated tests and Docker Compose infrastructure.

Network Monitoring and Incident System

IT Operations / Infrastructure | [Live demo](#) | [GitHub](#)

- Built a FastAPI platform for 5 simulated services with latency tracking, a configurable 3-consecutive-failure threshold, automatic incident creation and resolution on recovery.
- Added incident timelines, service-health metrics, PostgreSQL and Docker support, automated tests and optional integration with the AI Helpdesk API.

Document Verification and Trust System

Security / Cryptography | [Live demo](#) | [GitHub](#)

- Built a FastAPI platform using SHA-256 fingerprints for exact-copy verification and audit logging without storing original document contents.
- Implemented Ed25519-signed JSON/PDF receipts, QR-based public verification, PostgreSQL and Docker support, automated tests and an interactive GitHub Pages demo.

Australian Job Intelligence Dashboard

Data Analytics / NLP | [Live demo](#) | [GitHub](#)

- Built a job analytics platform using a curated Australian dataset of 240 records, 32 tracked technical skills and 10 locations, with interactive filters and Chart.js visualisations.
- Implemented CV-to-role Technical Skill Coverage using deterministic skill normalisation, a FastAPI/PostgreSQL backend, Docker setup, tests and a GitHub Pages demo.

ADDITIONAL

Registration: Registration to Work with Vulnerable People (Tasmania).

Work rights: Currently hold a Student visa, with work subject to visa conditions. Plan to apply for a Temporary Graduate visa (subclass 485) after course completion; future full-time work rights subject to visa grant and conditions.

References: Available on request.